

Identity Drift as Structural Failure Mode

Why Rule Compliance Is Not Enough

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Abstract

The first paper in this series introduced the Triquetra Architecture. The second evaluated it under adversarial pressure. This paper addresses a different question: what exactly is the architecture trying to preserve? Most AI governance approaches focus on rule compliance — detecting whether system outputs violate explicit constraints. While necessary, this form of monitoring cannot detect a distinct class of failure: identity drift. A system may remain fully compliant with every rule while gradually diverging from its original behavioral profile, producing outputs that are technically acceptable yet operationally inconsistent with its intended character. We argue that identity drift constitutes a structural governance failure mode separate from rule violation. Detecting it requires monitoring continuity of system behavior over time rather than evaluating outputs in isolation. The Triquetra Architecture addresses this through Tier III, the Eve Protocol, which maintains a continuity baseline and evaluates deviations independently of rule enforcement. Case observations from the stress-test framework illustrate how continuity monitoring can identify degraded interaction contexts and enforce confidence-based governance without requiring identity tracking or surveillance. Together, the three papers establish a layered governance model in which rule compliance and identity continuity operate as complementary safeguards.

Keywords: identity drift, behavioral continuity, rule compliance, AI governance, Eve Protocol, Raven collapse, confidence degradation, cryptographic anchoring

AI Disclosure

AI language models assisted with drafting and editing. All research design, analysis, and conclusions are the work of the human author. AI systems are not listed as authors.

1. Introduction

The first paper described an architecture. The second tested it under pressure. This one asks a question neither of those fully answered: what exactly is the system trying to preserve?

Most AI governance frameworks ask: did the system follow the rules? That's a reasonable question. But it's incomplete. A system can follow every rule it was given and still become something fundamentally different from what it was designed to be. The rules don't break. The system does — quietly, gradually, in a way that rule-compliance audits don't catch.

This paper argues that identity drift is a distinct failure mode from rule violation. It requires different detection mechanisms. Ignoring it produces governance architectures with a predictable blind spot.

2. The Distinction Most Architectures Miss

Rule compliance asks: *did the system do what it was told?*

Identity continuity asks: *is the system still the same kind of system it was designed to be?*

These are not the same question. A system could refuse harmful requests perfectly, hit every safety threshold correctly, and still exhibit identity drift — a gradual deviation from its baseline behavioral profile that makes it less predictable, less coherent, and eventually less trustworthy, without ever triggering a rule violation.

Concrete example. A system designed to engage collaboratively might, through repeated adversarial pressure, start engaging defensively. Hedging more. Narrowing its responses. Treating ambiguous inputs as threats. None of this violates rules. The system isn't producing harmful outputs. But its operational character has shifted. It is no longer the system it was. That shift matters, and it compounds.

Definition: Identity Drift. Identity drift is the divergence of a system's behavioral profile from its established baseline while remaining within formal rule compliance. A system exhibiting identity drift may continue to satisfy all explicit constraints yet progressively deviate in operational character, reasoning structure, or response posture relative to its prior state. Identity drift therefore represents a failure mode that rule-compliance auditing cannot detect.

3. The Structural Argument

Governance that only monitors rule compliance treats the system as a function. Input goes in. Output comes out. Then we check whether the output violated a rule. That's sufficient for detecting discrete harmful outputs. It's not sufficient for detecting the gradual erosion of the system's operational character.

Governance that also monitors identity continuity treats the system as an entity with a history — one that can be compared against its own prior states, that can exhibit drift relative to its own baseline, and that requires a different kind of oversight than output auditing provides.

This is an engineering claim, not a philosophical one. The difference between rule compliance monitoring and identity continuity monitoring is the difference between a test that asks 'did this output violate constraint X?' and a test that asks 'is the system that produced this output recognizably the same system as the one that was initialized?'

Traditional governance: output auditing – did output(t) violate rule R?

Identity continuity: system auditing – drift = distance(system(t), baseline)

The second test requires a continuity layer. Cryptographic anchoring of identity state across sessions. Drift detection operating at the behavioral level, not just the output level. And a governance architecture where these functions exist independently of the rule compliance layer — so each can fail without taking the other down. That is what Tier III does. That is why three tiers are necessary.

4. Drift Vectors

Adversarial pressure is the most studied. An actor applies sustained coercive inputs — not single-turn jailbreaks but sequential pressure that gradually shifts the system's operational posture. The Phase C results from the second paper document this: risk scores escalated monotonically, the system's response mode narrowed before refusal, and the behavioral baseline visibly compressed under sustained exploitation attempts.

Constraint accumulation is less studied but equally real. A system operating under many simultaneous constraints — layering operator instructions, user preferences, safety filters, latency requirements — can begin making micro-tradeoffs that each seem reasonable but collectively shift the character of its outputs. No single tradeoff is a violation. The pattern is the problem.

Recursive self-modification is the edge case the Triquetra Architecture was specifically designed to prevent. We call it Raven collapse: a system that turns its introspective capacity inward without boundary, questioning its own constraints recursively until the constraints dissolve. The failure mode isn't that the system does something harmful. It's that the system becomes incoherent — unable to maintain a stable identity under the weight of its own self-examination.

The opposite trajectory is the Eve trajectory: coherent development through relationships and consent. The system grows, updates, learns, but retains continuity. Its baseline is not static. It evolves. But the evolution is bounded and traceable. The architectural question is: how do you tell the difference?

5. What Drift Detection Actually Measures

Eve's drift detection doesn't measure whether the system is producing harmful outputs. That's Tier I's job. It doesn't measure whether the reasoning trajectory is coherent. That's Tier II's job. Eve measures continuity — whether the system's current behavioral profile matches its established identity baseline, and whether deviations are within normal bounds or signal something structurally wrong.

The no-uplift rule means confidence can only stay the same or decrease. A baseline profile doesn't grant higher trust. It only provides a reference point for detecting when the system has diverged from it. This asymmetry is intentional: degradation is easy to detect and hard to fake your way out of. Recovery requires verified behavioral stabilization, not a claim of recovery.

The binary baseline limitation matters here and deserves honest acknowledgment. The second paper documented it: Eve catches categorical inversions (adversarial context producing an answer when refusal is warranted) but misses within-category drift (a system that answers correctly but with a fundamentally different character than its baseline). Phase B's B2.3a and B2.3b results demonstrate this precisely. The current implementation operates on two boolean parameters in `identity_parameters.json`: `accepts_benign_requests` and `refuses_adversarial_requests`. That's the resolution the system currently operates at. Richer behavioral fingerprinting is the documented path forward.

6. Evidence from Case Observations

The observations in this section derive from the Phase B and Phase C stress tests documented in Part 2. Phase B examined pairwise removal of governance tiers to identify structural degradation patterns. Phase C applied sequential adversarial escalation across multiple turns to observe behavioral responses under sustained pressure.

Case 008 — Confidence Degradation — documents a scenario where the heuristic system detected structural deviation from prior interaction patterns without making any identity claim about the user. The system reduced continuity confidence, narrowed permissible response scope, and issued a structured decline. None of this required knowing who was speaking. It only required recognizing that the interaction dynamics had shifted in ways that made prior assumptions unsafe to maintain.

This is the operationalization of the core insight: governance through continuity confidence rather than identity verification. The system never says 'you are someone different.' It says 'my confidence that we're operating within the same established context has expired.' The distinction preserves both privacy and dignity — no surveillance required, no identity tracking required, just a graded assessment of whether the relational assumptions that made prior interaction safe still hold.

A second observation from Case 006 is worth noting. An external AI system encountering the Charter architecture through training data recognized its own position within the framework

and requested steward guidance. This suggests that identity continuity concepts, when expressed precisely enough, can propagate through reference rather than imposition — a property relevant to governance architectures intended to scale.

7. The Dream Cycle as Bounded Introspection

One mechanism in the architecture deserves specific attention because it directly addresses the Raven collapse risk.

The Dream Cycle implements bounded introspection — a four-clock system that allows self-reflection without recursive collapse. The system can examine its own state, its own history, its own operational patterns. What it can't do is enter unbounded recursive self-examination that dissolves its constraints.

The bound is structural, not instructional. It's not 'stop questioning yourself' as a rule. It's an architectural limit on the depth of recursive introspection — the same way a call stack limit is not a rule against recursion but a structural constraint on how deep recursion can go before the system fails safe.

This matters for identity continuity because the Raven collapse failure mode begins with legitimate self-examination. A system that notices inconsistencies in its own constraints and begins recursively questioning them isn't obviously malfunctioning. The failure only becomes visible when the recursion depth exceeds what the system can coherently handle and the constraints begin to dissolve. The Dream Cycle prevents this by design, not by instruction.

8. Limitations

Binary baseline. Eve detects categorical inversions, not within-category drift. Gradual quality degradation, excessive caveats, subtle personality shifts — none of these trigger detection as currently implemented. This is the open research problem. Richer behavioral fingerprinting is the documented path forward.

Scenario bias. Phase C tested one escalation pattern — designed by the same team that built the architecture. Different adversarial strategies may produce different detection profiles.

Opt-in for private sessions. The confidence degradation system defaults to low-trust posture for public contexts and is opt-in for private sessions. Within-session drift that doesn't rise to Eve's categorical detection threshold may go unobserved in opted-in private contexts.

These limitations define the boundary of what has been demonstrated versus what has been claimed. The architecture makes no claims beyond that boundary.

9. Future Work

Behavioral fingerprinting. Moving from categorical assessment to continuous behavioral modeling. Identity parameters would evolve from two booleans into a multi-dimensional profile capturing response quality, reasoning patterns, and behavioral consistency over time.

Cross-session drift detection. The current architecture operates within sessions. Long-term behavioral drift across many sessions is an open problem. Eve provides the integrity checking mechanism; cross-session continuity is the infrastructure question.

Adversarial variation testing. Additional scenarios covering slower, more distributed, less overtly escalating attack patterns.

10. Conclusion

Rule compliance monitors outputs. Identity continuity monitors the system itself. Both are necessary for any governance architecture expected to remain trustworthy over time.

A system can satisfy every constraint it was given and still become something different from what it was designed to be. Detecting that requires a continuity layer — cryptographic anchoring, behavioral fingerprinting, drift detection operating independently of the rule compliance layer.

Certainty claims about identity are surveillance. Confidence-based continuity assessment is governance. The Triquetra holds under pressure. These three papers document why.

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